```
===== MENU =====
1. Load Dataset
2. Explore Data
3. Clean Data
4. Numpy Operations
5. Mathematical Operations
6. Search/Sort/Filter
7. Aggregation
8. Statistical Analysis
9. Pivot Table
10. Matplotlib Visualization
11. Seaborn Visualization
12. Save Visualization
0. Exit
Enter choice: 1
Enter CSV file path: D:\PYTHON\Project-9 Pandas Analyzer & Data Visualization\product_sales_dataset_final.csv
Data loaded successfully

===== MENU =====
1. Load Dataset
2. Explore Data
3. Clean Data
4. Numpy Operations
5. Mathematical Operations
6. Search/Sort/Filter
7. Aggregation
8. Statistical Analysis
9. Pivot Table
10. Matplotlib Visualization
11. Seaborn Visualization
12. Save Visualization
0. Exit
Enter choice: 2
First 5 rows:
   Order_ID Order_Date   Customer_Name         City        State  Region        Country            Category      Sub_Category       Product_Name  Quantity  Unit_Price  Revenue  Profit
0         1   08-23-23     Bianca Brown       Jackson  Mississippi   South  United States         Accessories  Small Electronics         Phone Case         3      201.01   603.03  221.49
1         2   12-20-24    Jared Edwards  Grand Rapids     Michigan  Centre  United States         Accessories  Small Electronics     Charging Cable         4       74.30   297.20   97.09
2         3   01-29-24     Susan Valdez   Minneapolis    Minnesota  Centre  United States  Clothing & Apparel         Sportswear   Nike Air Force 1         1       68.19    68.19   25.47
3         4   11-29-24    Tina Williams   Tallahassee      Florida   South  United States  Clothing & Apparel         Sportswear  Adidas Tracksuit         3      209.64   628.92  231.38
4         5   09-21-23  Catherine Gordon    Baltimore     Maryland    East  United States         Accessories               Bags           Backpack         1      216.63   216.63   42.46
<class 'pandas.DataFrame'>
```

```
<class 'pandas.DataFrame'>
RangeIndex: 200000 entries, 0 to 199999
Data columns (total 14 columns):
 #   Column         Non-Null Count   Dtype
---  ------         --------------   -----
 0   Order_ID       200000 non-null  int64
 1   Order_Date     200000 non-null  str
 2   Customer_Name  200000 non-null  str
 3   City           200000 non-null  str
 4   State          200000 non-null  str
 5   Region         200000 non-null  str
 6   Country        200000 non-null  str
 7   Category       200000 non-null  str
 8   Sub_Category   200000 non-null  str
 9   Product_Name   200000 non-null  str
 10  Quantity       200000 non-null  int64
 11  Unit_Price     200000 non-null  float64
 12  Revenue        200000 non-null  float64
 13  Profit         200000 non-null  float64
dtypes: float64(3), int64(2), str(9)
memory usage: 21.4 MB
Data Info:
 None
Data Description:
            Order_ID       Quantity     Unit_Price        Revenue         Profit
count  200000.000000  200000.000000  200000.000000  200000.000000  200000.000000
mean   100000.500000       1.854000     382.855615     712.038725     157.743041
std     57735.171256       1.100536     276.870235     742.471556     155.689581
min         1.000000       1.000000      17.030000      17.030000       3.920000
25%     50000.750000       1.000000     162.760000     229.187500      59.210000
50%    100000.500000       1.000000     303.545000     464.880000     109.530000
75%    150000.250000       2.000000     562.252500     881.302500     199.402500
max    200000.000000      11.000000    1432.000000    9014.250000    2763.720000
```

```
===== MENU =====
1. Load Dataset
2. Explore Data
3. Clean Data
4. Numpy Operations
5. Mathematical Operations
6. Search/Sort/Filter
7. Aggregation
8. Statistical Analysis
9. Pivot Table
10. Matplotlib Visualization
11. Seaborn Visualization
12. Save Visualization
0. Exit
Enter choice: 3
Missing values removed

===== MENU =====
1. Load Dataset
2. Explore Data
3. Clean Data
4. Numpy Operations
5. Mathematical Operations
6. Search/Sort/Filter
7. Aggregation
8. Statistical Analysis
9. Pivot Table
10. Matplotlib Visualization
11. Seaborn Visualization
12. Save Visualization
0. Exit
Enter choice: 4
First 5 elements: [603.03 297.2   68.19 628.92 216.63]
Single element: 603.03
```

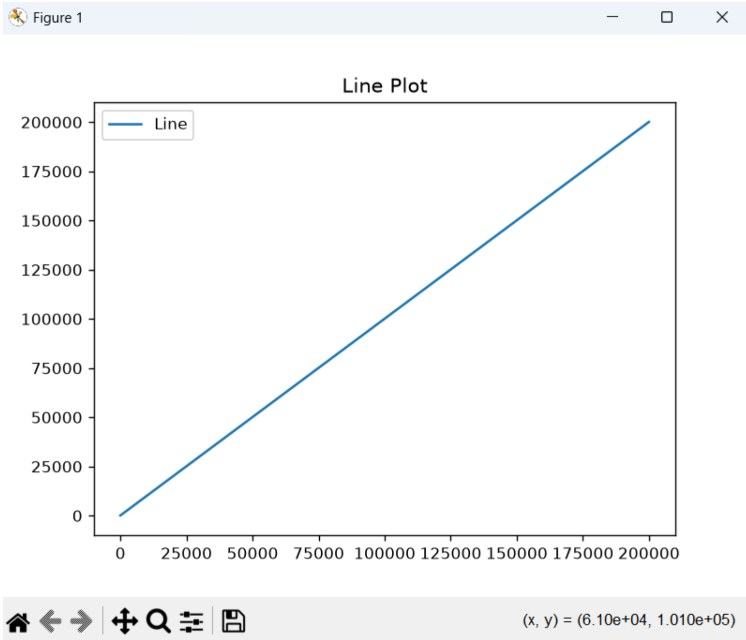
```
===== MENU =====
1. Load Dataset
2. Explore Data
3. Clean Data
4. Numpy Operations
5. Mathematical Operations
6. Search/Sort/Filter
7. Aggregation
8. Statistical Analysis
9. Pivot Table
10. Matplotlib Visualization
11. Seaborn Visualization
12. Save Visualization
0. Exit
Enter choice: 5
   Order_ID Order_Date    Customer_Name          City        State  Region        Country  ...       Sub_Category      Product_Name Quantity  Unit_Price  Revenue  Profit      Sum
0         1   08-23-23     Bianca Brown       Jackson  Mississippi   South  United States  ...  Small Electronics        Phone Case        3      201.01   603.03  221.49  1029.53
1         2   12-20-24    Jared Edwards  Grand Rapids     Michigan  Centre  United States  ...  Small Electronics    Charging Cable        4       74.30   297.20   97.09   474.59
2         3   01-29-24     Susan Valdez   Minneapolis    Minnesota  Centre  United States  ...         Sportswear  Nike Air Force 1        1       68.19    68.19   25.47   165.85
3         4   11-29-24    Tina Williams   Tallahassee      Florida   South  United States  ...         Sportswear  Adidas Tracksuit        3      209.64   628.92  231.38  1076.94
4         5   09-21-23  Catherine Gordon     Baltimore     Maryland    East  United States  ...               Bags          Backpack        1      216.63   216.63   42.46   481.72

[5 rows x 15 columns]
```

```
===== MENU =====
1. Load Dataset
2. Explore Data
3. Clean Data
4. Numpy Operations
5. Mathematical Operations
6. Search/Sort/Filter
7. Aggregation
8. Statistical Analysis
9. Pivot Table
10. Matplotlib Visualization
11. Seaborn Visualization
12. Save Visualization
0. Exit
Enter choice: 6
Column: City
Value: Jackson
        Order_ID Order_Date    Customer_Name          City        State Region        Country  ...            Sub_Category          Product_Name Quantity  Unit_Price   Revenue   Profit       Su
0              1   08-23-23     Bianca Brown       Jackson  Mississippi  South  United States  ...       Small Electronics            Phone Case        3      201.01    603.03   221.49   1029.5
62            63   01-13-24 Brianna Hamilton  Jacksonville      Florida  South  United States  ...   Wearable Accessories           Watch Strap        2      121.49    242.98    65.79    495.2
209          210   10-08-24   Tyler Thompson  Jacksonville      Florida  South  United States  ...                 Bedding  Tempur-Pedic Mattress        2      579.63   1159.26   189.81   2140.7
223          224   10-14-23 Jacqueline Adams  Jacksonville      Florida  South  United States  ...             Kitchenware           Instant Pot        3      531.89   1595.67   210.23   2564.7
290          291   05-16-24      Luis Harris  Jacksonville      Florida  South  United States  ...               Men's Wear          Levi's Jeans        1      213.45    213.45    80.97    799.8
...          ...        ...              ...           ...          ...    ...            ...  ...                     ...                   ...      ...         ...       ...      ...       ..
199608    199609   03-22-24      Mary Arroyo  Jacksonville      Florida  South  United States  ...               Furniture       Dining Table Set        3      174.87    524.61   144.49 200455.9
199609    199610   05-22-23    Daniel Graham  Jacksonville      Florida  South  United States  ...                 Bedding  Tempur-Pedic Mattress        1      802.91    802.91   126.77 201343.5
199696    199697   12-31-24  Morgan Martinez  Jacksonville      Florida  South  United States  ...                 Bedding      Brooklinen Sheets        1      224.43    224.43    30.36 200177.2
199887    199888   09-19-23   Sandra Johnson       Jackson  Mississippi  South  United States  ...                 Bedding      Brooklinen Sheets        2      908.00   1816.00   337.41 202951.4
199930    199931   05-26-23       Wendy Neal  Jacksonville      Florida  South  United States  ...                    Bags              Tote Bag        1      238.06    238.06    86.83 200494.9

[2140 rows x 15 columns]
Sort column: Profit
Order_ID > 199995
        Order_ID Order_Date   Customer_Name           City          State  Region        Country  ...           Sub_Category    Product_Name Quantity  Unit_Price   Revenue   Profit        Sum
199995    199996   08-15-23 William Jackson         Boston  Massachusetts    East  United States  ...                Storage    Storage Rack        4      254.42   1017.68   334.72  201606.82
199996    199997   10-17-23   Sharon Ferrell       Bismarck   North Dakota  Centre  United States  ...      Small Electronics  Charging Cable        1      237.04    237.04    46.53  200518.61
199997    199998   12-03-23     Katie Rivera       Santa Fe     New Mexico    West  United States  ...   Wearable Accessories      Sunglasses        2      106.83    213.66   102.57  200423.06
199998    199999   12-08-23    Lisa Sullivan  New York City       New York    East  United States  ...             Women's Wear     Zara Blouse        2      353.01    706.02   288.74  201348.77
199999    200000   12-13-24  Jessica Mueller        Augusta          Maine    East  United States  ...               Men's Wear      GAP Hoodie        1      216.68    216.68    59.10  200493.46

[5 rows x 15 columns]
```

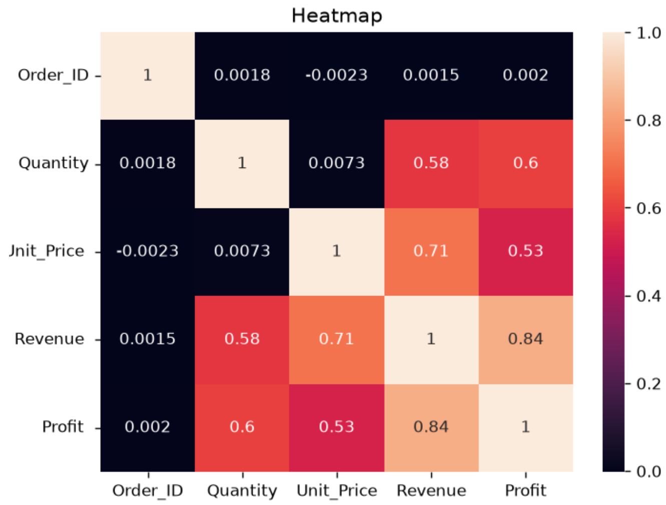
```
===== MENU =====
1. Load Dataset
2. Explore Data
3. Clean Data
4. Numpy Operations
5. Mathematical Operations
6. Search/Sort/Filter
7. Aggregation
8. Statistical Analysis
9. Pivot Table
10. Matplotlib Visualization
11. Seaborn Visualization
12. Save Visualization
0. Exit
Enter choice: 7
Sum:
 Order_ID      2.000010e+10
Quantity      3.708000e+05
 Unit_Price    7.657112e+07
 Revenue       1.424077e+08
 Profit        3.154861e+07
Sum           2.025100e+10
dtype: float64
Mean:
 Order_ID      100000.500000
Quantity           1.854000
 Unit_Price      382.855615
 Revenue         712.038725
 Profit          157.743041
Sum           101254.991380
dtype: float64
Count:
 Order_ID      200000
Quantity      200000
 Unit_Price    200000
 Revenue       200000
 Profit        200000
Sum           200000
dtype: int64
```

```
===== MENU =====
1. Load Dataset
2. Explore Data
3. Clean Data
4. Numpy Operations
5. Mathematical Operations
6. Search/Sort/Filter
7. Aggregation
8. Statistical Analysis
9. Pivot Table
10. Matplotlib Visualization
11. Seaborn Visualization
12. Save Visualization
0. Exit
Enter choice: 8
Standard Deviation:
 Order_ID      57735.171256
Quantity          1.100536
 Unit_Price     276.870235
 Revenue        742.471556
 Profit         155.689581
Sum           57746.196780
dtype: float64
Variance:
 Order_ID      3.333350e+09
Quantity      1.211180e+00
 Unit_Price    7.665713e+04
 Revenue       5.512640e+05
 Profit        2.423925e+04
Sum           3.334623e+09
dtype: float64
Percentiles:
        Order_ID  Quantity  Unit_Price  Revenue   Profit          Sum
0.25    50000.75       1.0    162.7600  229.1875   59.2100   51282.5550
0.50   100000.50       1.0    303.5450  464.8800  109.5300  101252.9300
0.75   150000.25       2.0    562.2525  881.3025  199.4025  151280.1075
```

```
===== MENU =====
1. Load Dataset
2. Explore Data
3. Clean Data
4. Numpy Operations
5. Mathematical Operations
6. Search/Sort/Filter
7. Aggregation
8. Statistical Analysis
9. Pivot Table
10. Matplotlib Visualization
11. Seaborn Visualization
12. Save Visualization
0. Exit
Enter choice: 10

1.Bar 2.Line 3.Scatter 4.Pie 5.Histogram 6.Stack Plot
Choose plot: 2
```




```
===== MENU =====
1. Load Dataset
2. Explore Data
3. Clean Data
4. Numpy Operations
5. Mathematical Operations
6. Search/Sort/Filter
7. Aggregation
8. Statistical Analysis
9. Pivot Table
10. Matplotlib Visualization
11. Seaborn Visualization
12. Save Visualization
0. Exit
Enter choice: 11
```




```
===== MENU =====
1. Load Dataset
2. Explore Data
3. Clean Data
4. Numpy Operations
5. Mathematical Operations
6. Search/Sort/Filter
7. Aggregation
8. Statistical Analysis
9. Pivot Table
10. Matplotlib Visualization
11. Seaborn Visualization
12. Save Visualization
0. Exit
Enter choice: 12
Enter file name (with .png or .jpg): line.png
Plot saved
```